## MISC

**Quadratic formula**:

$$ax^2+bx+c=0 \Rightarrow x_{1,2}=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

**Monotonically in-/decreasing**:

$$\forall x.\ (f'(x)>0)\Rightarrow a>b \Leftrightarrow f(a)>f(b)$$
$$\forall x.\ (f'(x)<0)\Rightarrow a>b \Leftrightarrow f(a)<f(b)$$

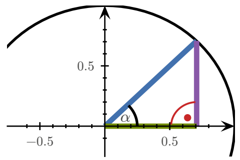
**RSS**:

$$\mathrm{RSS}=\sum_i (y_i-f(x_i))^2$$

## TRIGONOMETRY

| $x$ | $0$ | $30°={}^{\pi}/_{6}$ | $45°={}^{\pi}/_{4}$ | $60°={}^{\pi}/_{3}$ | $90°={}^{\pi}/_{2}$ |
|---|---|---|---|---|---|
| $\sin(x)$ | $0$ | ${}^{1}/_{2}$ | ${}^{\sqrt{2}}/_{2}$ | ${}^{\sqrt{3}}/_{2}$ | $1$ |
| $\cos(x)$ | $1$ | ${}^{\sqrt{3}}/_{2}$ | ${}^{\sqrt{2}}/_{2}$ | ${}^{1}/_{2}$ | $0$ |
| $\tan(x)$ | $0$ | ${}^{1}/_{\sqrt{3}}$ | $1$ | $\sqrt{3}$ | |

$$\tan(x)=\frac{\sin(x)}{\cos(x)}$$
$$\sin(2x)=2\sin(x)\cos(x)$$
$$\sin^2(x)+\cos^2(x)=1$$
$$1, \sin(\alpha), \cos(\alpha)$$




## VECTORS

| *Term* | *Definition* |
|---|---|
| Norm/Length | $\|v\|=\sqrt{\sum_{i=1}^n v_i^2}=\sqrt{v^\top v}$<br>$\nabla\|v\|=\frac{1}{\|v\|}v^\top$<br>$\nabla\frac{1}{\|v\|}=-\frac{1}{\|v\|^2}\nabla\|v\|=-\frac{1}{\|v\|^3}v^\top$ |
| Distance | $d(u,v)=\|u-v\|$<br>$\nabla d(u,v)=\frac{1}{d(u,v)}((u-v)^\top(v-u)^\top)$ |
| Scalar product | $u\bullet v=u^\top v=\sum_k u_k v_k$<br>$\nabla(u\bullet v)=(v^\top\ u^\top)$ |

## MATRICES

**Affine transformation**:

$$M\in\mathbb{R}^{m\times n},\quad A_{b,M}:\begin{cases}\mathbb{R}^m\to\mathbb{R}^n\\ x\ \mapsto b+M\cdot x\end{cases}$$

**Determinant**:

$$\det(a)=a \qquad \det\begin{pmatrix}a&b\\c&d\end{pmatrix}=ad-cb$$

$$\det\begin{pmatrix}a_{11}&a_{12}&a_{13}\\a_{21}&a_{22}&a_{23}\\a_{31}&a_{32}&a_{33}\end{pmatrix}=a_{11}a_{22}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{32}$$
$$-a_{31}a_{22}a_{13}-a_{32}a_{23}a_{11}-a_{33}a_{21}a_{12}$$

$$\det(A\cdot B)=\det(A)\cdot\det(B)\qquad \det(\mathbb{1})=1$$
$$\det(A^{-1})=\frac{1}{\det(A)}\qquad \det(A^\top)=\det(A)$$

**Inverting**:

$$A=\begin{pmatrix}a&b\\c&d\end{pmatrix}\Rightarrow A^{-1}=\frac{1}{ad-cb}\begin{pmatrix}d&-b\\-c&a\end{pmatrix}$$

**Transposing**:

$$\begin{pmatrix}1&0\\0&2\\3&1\end{pmatrix}^\top=\begin{pmatrix}1&0&3\\0&2&1\end{pmatrix}\qquad (A^\top)^\top=A\qquad (A+B)^\top=A^\top+B^\top$$
$$\begin{pmatrix}1\\4\\5\end{pmatrix}^\top=(1\ 4\ 5)\qquad (\lambda A)^\top=\lambda A^\top\qquad (AB)^\top=B^\top A^\top$$

**Matrix multiplication**:

$$\begin{pmatrix}2&3&1\\4&-1&7\end{pmatrix}\cdot\begin{pmatrix}6&4\\1&0\\8&9\end{pmatrix}$$
$$=\begin{pmatrix}2\cdot6+3\cdot1+1\cdot8 & 2\cdot4+3\cdot0+1\cdot9\\ 4\cdot6+-1\cdot1+7\cdot8 & 4\cdot4+-1\cdot0+7\cdot9\end{pmatrix}=\begin{pmatrix}23&17\\79&79\end{pmatrix}$$

## PROPERTIES

$$A\cdot A^{-1}=\mathbb{1}\qquad \ker A=\{v\mid Av=0\}$$
$$A+B=B+A\qquad A\cdot B\neq B\cdot A$$
$$A+(B+C)=(A+B)+C\qquad A(BC)=(AB)C$$
$$A(B+C)=AB+AC\qquad (A+B)C=AC+BC$$

## INDICES

$$A\in\mathbb{R}^{n\times m}, B\in\mathbb{R}^{m\times r},\qquad A\cdot B\in\mathbb{R}^{n\times r}$$
$$(A\cdot B)_{ij}=\sum_{k=1}^m A_{ik}B_{kj}\quad=\sum_k A_{ik}B_{kj}$$
$$((A\cdot B)^\top)_{ij}=(A\cdot B)_{ji}\quad=\sum_k A_{jk}B_{ki}$$
$$(A\cdot B^\top)_{ij}=\sum_k A_{ik}B^\top_{kj}\quad=\sum_k A_{ik}B_{jk}$$
$$a^\top\cdot B\cdot b=\sum_{ij}(a^\top)_i B_{ij}b_j\quad=\sum_{ij}a_iB_{ij}b_j$$
$$(B\cdot a)(C\cdot a)=\sum_i(B\cdot a)_i(C\cdot a)_i$$

### ADDITION/SUBTRACTION

$$A\in\mathbb{R}^{n\times m}, B\in\mathbb{R}^{n\times m}$$
$$(A+B)_{ij}=A_{ij}+B_{ij}$$
$$(A-B)_{ij}=A_{ij}-B_{ij}$$

## KRONECKER DELTA

$$\delta_{ij}=(\vec{e}_i)_j=\begin{cases}1 &, i=j\\ 0 &, \text{otherwise}\end{cases}$$
$$(E_{rst})_{ijk}=\delta_{ri}\delta_{sj}\delta_{tk}$$

## BASIS VECTORS

$$\mathbb{R}^3$$
$$e_1=\begin{pmatrix}1\\0\\0\end{pmatrix}, e_2=\begin{pmatrix}0\\1\\0\end{pmatrix}, e_3=\begin{pmatrix}0\\0\\1\end{pmatrix}$$

## UNIT MATRICES

$$\mathbb{R}^2$$
$$E_{11}=\begin{pmatrix}1&0\\0&0\end{pmatrix}, E_{12}=\begin{pmatrix}0&1\\0&0\end{pmatrix}, E_{21}=\begin{pmatrix}0&0\\1&0\end{pmatrix}, E_{22}=\begin{pmatrix}0&0\\0&1\end{pmatrix}$$

## AFFINE TRANSFORMATIONS

$$A_{a,M}(x)=a+M\cdot x$$
$$L_A(\vec{0})=\vec{0}$$

## DERIVATIVES

| | | | |
|---|---|---|---|
| $1$ | $\to 0$ | $x$ | $\to 1$ |
| $x^2$ | $\to 2x$ | $x^n$ | $\to nx^{n-1}$ |
| $\frac{1}{x}$ | $\to -\frac{1}{x^2}$ | $\sqrt{x}$ | $\to \frac{1}{2\sqrt{x}}$ |
| $e^x$ | $\to e^x$ | $e^{-x}$ | $\to -e^{-x}$ |
| $\ln(x)$ | $\to \frac{1}{x}, x>0$ | $\ln(y\cdot x)$ | $\to \frac{1}{x}, x>0$ |
| $a^x$ | $\to \ln(a)\cdot a^x, a>0, a\neq 1$ | $\log_b(x)$ | $\to \frac{1}{\ln(b)\cdot x}$ |
| $\sin(x)$ | $\to \cos(x)$ | $\sin(2x)$ | $\to 2\cos(2x)$ |
| $\cos(x)$ | $\to -\sin(x)$ | $\cos(ax)$ | $\to -a\sin(ax)$ |
| $\tan(x)$ | $\to 1+\tan^2(x)=\frac{1}{\cos^2(x)}$ | $\operatorname{arccot}(x)$ | $\to -\frac{1}{1+x^2}$ |
| $\cot(x)$ | $\to -\frac{1}{\sin^2(x)}$ | $\arcsin(x)$ | $\to \frac{1}{\sqrt{1-x^2}}$ |
| $\arccos(x)$ | $\to -\frac{1}{\sqrt{1-x^2}}$ | $\arctan(x)$ | $\to \frac{1}{1+x^2}$ |

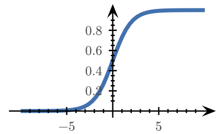
| *Case* | *Rule* |
|---|---|
| Addition | $(f(x)+g(x))'=f'(x)+g'(x)$ |
| Multiplication | $(\alpha\cdot f(x))'=\alpha\cdot f'(x)$ |
| Product rule | $(f\cdot g)'=f'\cdot g+f\cdot g'$<br>$(f\cdot g\cdot h)'=f'\cdot g\cdot h+f\cdot g'\cdot h+f\cdot g\cdot h'$ |
| Chain rule | $(f(g(x)))'=f'(g(x))\cdot g'(x)$ |
| Quotient rule | $\left(\frac{f}{g}\right)'=\frac{f'g-fg'}{g^2}$ |

## FUNCTIONS OF SEVERAL VARIABLES

| *Term* | *Definition* |
|---|---|
| Real valued function | $R\subset\mathbb{R}$ |
| Vector valued function | $R\subset\mathbb{R}^m$ |
| A function of $n$ variables | $D\subset\mathbb{R}^n$ |
| $n$-dimensional curve | $c:\mathbb{R}\to\mathbb{R}^n$ |
| $n$-dimensional hyper-surface | $s:\mathbb{R}^n\to\mathbb{R}$ |
| Reparametrization | $c(t)\to d(s)=c(h(s))$ |
| Softmax function | $\sigma:\begin{cases}\mathbb{R}^n\to[0;1]^n\\ x_i\ \mapsto e^{x_i}\left(\sum_{j=1}^n e^{x_j}\right)^{-1}\end{cases}$ |

**Sigmoid function**:

$$\sigma(x):\mathbb{R}\to[0;1]=\frac{1}{1+e^{-x}}$$




## DERIVATIVES

| *Function* | *Derivative* |
|---|---|
| $f:\begin{cases}\mathbb{R}\to\mathbb{R}^n\\ x\mapsto y\end{cases}$ | $f':\begin{cases}\mathbb{R}\to\mathbb{R}^n\\ x\mapsto y\end{cases}=\begin{pmatrix}\frac{d}{dx}f_1(x)\\ \frac{d}{dx}f_2(x)\\ \vdots\\ \frac{d}{dx}f_n(x)\end{pmatrix}$ |
| $f:\begin{cases}\mathbb{R}^n\to\mathbb{R}\\ x\ \mapsto y\end{cases}$ | $f':\begin{cases}\mathbb{R}^n\to\mathbb{R}^{1\times n}\\ x\ \mapsto y\end{cases}=\nabla f$<br>$=\left(\frac{\partial}{\partial x_1}f(x),\frac{\partial}{\partial x_2}f(x),\dots,\frac{\partial}{\partial x_n}f(x)\right)$ |
| $f:\begin{cases}\mathbb{R}^n\to\mathbb{R}^m\\ x\ \mapsto y\end{cases}$ | $f':\begin{cases}\mathbb{R}^n\to\mathbb{R}^{m\times n}\\ x\ \mapsto y\end{cases}=J_f=\left.\frac{\partial f}{\partial x}\right\|_{x=x_0}$<br>$=\begin{pmatrix}\nabla f_1(x)\\ \vdots\\ \nabla f_m(x)\end{pmatrix}=\begin{pmatrix}\frac{\partial}{\partial x_1}f_1(x)&\dots&\frac{\partial}{\partial x_n}f_1(x)\\ \vdots&\ddots&\vdots\\ \frac{\partial}{\partial x_1}f_m(x)&\dots&\frac{\partial}{\partial x_n}f_m(x)\end{pmatrix}$ |

### PARTIAL VS TOTAL DERIVATIVE

***TODO:***
example FS2024

$$f(x,y)=x^2+2y$$
$$\frac{\partial f}{\partial x}=2x$$
$$\frac{df}{dx}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\cdot\frac{dy}{dx}=2x+2\cdot\frac{dy}{dx}\ \mid(\text{y dependent on x})$$

### GRADIENT VS TOTAL DERIVATIVE

$$f(x,y)=xye^y$$
$$x(t)=e^t$$
$$y(t)=t^2$$
$$\nabla f=(ye^y, xe^y+xye^y)\quad=e^y(y,x+xy)$$
$$\frac{d}{dt}f=e^y(y,x+xy)\cdot\frac{d}{dt}\begin{pmatrix}e^t\\t^2\end{pmatrix}\quad=\frac{d}{dt}\left(e^tt^2e^{t^2}\right)$$
$$=(2t+t^2+2t^3)e^{t^2+t}$$

## GRADIENTS

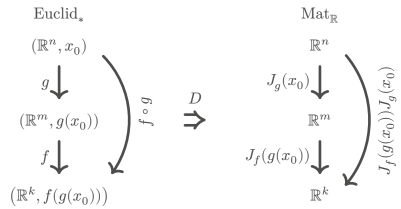
**Common gradients**:

$$f(x)=\|x\|^2\qquad\Rightarrow\nabla f=2x$$
$$f(x)=a^\top x+c\qquad\Rightarrow\nabla f=a$$
$$f(x)=x^\top Ax\qquad\Rightarrow\nabla f=(A+A^\top)x$$
$$f(x)=\frac{1}{2}x^\top Ax+b^\top x+c\Rightarrow\nabla f=\frac{A+A^\top}{2}x+b$$
$$=Ax+b\text{ if }A^\top=A$$
$$f(x)=g(Ax+b)\qquad\Rightarrow\nabla f=A^\top\nabla g(Ax+b)$$
$$g:\mathbb{R}^m\to\mathbb{R}$$

## JACOBIAN MATRIX

$$f:\mathbb{R}^n\to\mathbb{R}^m\qquad x_0\in\mathbb{R}^n$$
$$J_f(x_0):\mathbb{R}^{m\times n}\qquad J_f(x_0)_{ij}=\frac{\partial}{\partial x_j}f_i(x_0)$$
$$f:\mathbb{R}^2\to\mathbb{R}^2\qquad J_f(x,y)=\begin{pmatrix}\partial x_1&\partial y_1\\ \partial x_2&\partial y_2\end{pmatrix}$$

ty lukas <3




**Chain rule**:

$$f:\mathbb{R}^n\to\mathbb{R}^m,\ g:\mathbb{R}^m\to\mathbb{R}^k,\ h:\mathbb{R}^n\to\mathbb{R}^k=g\circ f$$
$$J_h=J_g(f(x))\cdot J_f(x)$$

**Common jacobians**:

$$f(x)=a+M\cdot x\Rightarrow J_f=M$$
$$f(x)=x\qquad\Rightarrow J_f=\mathbb{1}$$
$$f(x)=\|x\|^2\qquad\Rightarrow J_f=(2x)^\top$$

### LINEARISATION

$$f:\mathbb{R}^n\to\mathbb{R}^m,\ x_0\in\mathbb{R}^n$$
$$L_f(x)=f(x_0)+J_f(x_0)\cdot(x-x_0)$$

For finding better approximations for $x_0$:
1) Calculate $J_f$, then $J_f(x_0)$, then $L_f$
2) Find $x$ so that $L_f(x)\overset{!}{=}0$ (Gauss)

For calculating an approximate value at $x$:
1) $x_0$ = round $x$ to ints
2) Calculate $L_f(x)$ at $x_0$

## SECOND DERIVATIVE

$$\frac{\partial}{\partial x}\left(\frac{\partial}{\partial x}(f(x,y))\right)=\frac{\partial^2f(x,y)}{\partial x^2}=\partial_{xx}f(x,y)$$

## HESSIAN MATRIX

$$f:\mathbb{R}^n\to\mathbb{R},\quad x\in\mathbb{R}^n,\quad H(x)\in\mathbb{R}^{n\times n}$$
$$H(x,y)=\begin{pmatrix}f_{xx}&f_{xy}\\f_{yx}&f_{yy}\end{pmatrix}=J(\nabla f)$$
$$(H(x))_{ij}=\frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j}f(x)$$
$$\partial_i\partial_jf=\partial_j\partial_if$$

**Common hessians**:

$$f(x)=\frac{1}{2}x^\top Ax+b^\top x+c\Rightarrow H_f=\frac{A+A^\top}{2}$$
$$f(x)=g(Ax+b)\qquad\Rightarrow H_f=A^\top H_g(Ax+b)A$$
$$g:\mathbb{R}^m\to\mathbb{R}$$

### EXTREMAL POINTS

**Quadratic form**:

$$M\in\mathbb{R}^{n\times n}\qquad q_M(v)=v^\top Mv\qquad v\in\mathbb{R}^n$$
$$H=\frac{1}{2}(M+M^\top)\quad\forall v\in\mathbb{R}^n.\ (q_M(v)=q_H(v))$$

$v^\top Hv>0\Rightarrow c_0$ is a ***local minimum***, $H$ is ***positive definite***

$v^\top Hv\geq0\Rightarrow c_0$ is a ***local minimum*** or ***non-extremal***, $H$ is ***positive semi-definite***

$v^\top Hv<0\Rightarrow c_0$ is a ***local maximum***, $H$ is ***negative definite***

$v^\top Hv\leq0\Rightarrow c_0$ is a ***local maximum*** or ***non-extremal***, $H$ is ***negative semi-definite***

$v^\top Hv\neq0\Rightarrow c_0$ is a ***saddle point***, $H$ is ***indefinite***

**Finding stationary points**:
1) Calculate $\nabla f$, then $H_f$
2) Find stationary point(s) $x$ so that $\nabla f(x)\overset{!}{=}0$
3) Find type of point using technique of completing squares on $H_f(x)$

$$(v_1,v_2,v_3)\cdot\begin{pmatrix}h_{11}&h_{12}&h_{13}\\h_{21}&h_{22}&h_{23}\\h_{31}&h_{32}&h_{33}\end{pmatrix}\cdot\begin{pmatrix}v_1\\v_2\\v_3\end{pmatrix}$$
$$=(v_1,v_2,v_3)\begin{pmatrix}v_1h_{11}+v_2h_{12}+v_3h_{13}\\v_1h_{21}+v_2h_{22}+v_3h_{23}\\v_1h_{31}+v_2h_{32}+v_3h_{33}\end{pmatrix}$$
$$=v_1^2h_{11}+v_1v_2h_{12}+v_1v_3h_{13}$$
$$+v_1v_2h_{21}+v_2^2h_{22}+v_3v_2h_{23}$$
$$+v_1v_3h_{31}+v_2v_3h_{32}+v_3^2h_{33}$$
$$=v_1^2h_{11}+v_2^2h_{22}+v_3^2h_{33}+2v_1v_2h_{12}+2v_2v_3h_{23}+2v_1v_3h_{13}$$

**Technique of completing squares**:

$$\underbrace{x^2}_{\sqrt{\cdot}}-\underbrace{4xy}_{\div 2x}+1$$
$$=(x-2y)^2-(-2y)^2+1$$
$$=(x-2y)^2-4y^2+1$$

## HYPER-SURFACES

**Chain rule**:

$$f:\mathbb{R}^n\to\mathbb{R}^m,\ g:\mathbb{R}^m\to\mathbb{R},\ h:\mathbb{R}^n\to\mathbb{R}=g\circ f$$
$$\nabla h(x)=\nabla g(f(x))=\nabla g(f)|_{f=f(x)}\cdot\frac{\partial}{\partial x}f(x)$$

**Differentiation properties**:

$$f:\mathbb{R}^n\to\mathbb{R},\ g:\mathbb{R}^n\to\mathbb{R}$$
$$\nabla(f+g)=\nabla f+\nabla g$$
$$\nabla(f-g)=\nabla f-\nabla g$$
$$\nabla(f\cdot g)=g\nabla f+f\nabla g$$
$$\nabla\frac{f}{g}=\frac{g\nabla f-f\nabla g}{g^2}$$

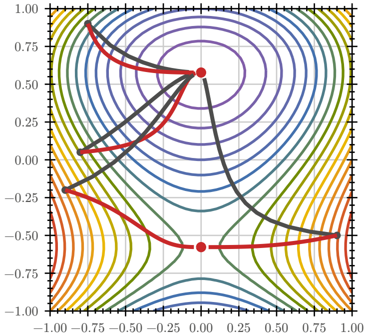
## COMMUTATIVE DIAGRAMS

$f:D\to M$
$g:N\to R$
$h:D\to R$
$=g\circ f$

$$\begin{tikzcd} D \arrow[r, "f"] \arrow[rr, bend left, "g\circ f"] & M \arrow[r, "g"] & R \end{tikzcd}$$

## GRADIENT DESCENT / NEWTON'S METHOD

$$GD\qquad x_{i+1}=x_i-\gamma\cdot\nabla f(x_i)$$
$$N\qquad x_{i+1}=x_i-\gamma\cdot\left(H_{f(x_i)}\right)^{-1}\cdot\nabla f(x_i)$$
$$\gamma=\text{step size}$$
$$\text{termination criterion }\varepsilon>|x_{i+1}-x_i|$$



## PROBABILITY THEORY

| *Term* | *Definition* |
| --- | --- |
| $\sim$ | Distributed |
| $\mathcal{N}(\mu,\sigma)$ | Normal distribution |
| $\mathrm{unif}(a,b)$ | Uniform distribution |
| $\mathbb{P}(E)$ | Probability measure |
| $f(x_i \mid \mu,\sigma)$ | Density |
| $l(\mu,\sigma)$ | Likelihood function $$\prod_{i=1}^{n} f(x_i \mid \mu,\sigma)$$ |
| $\ln(l(\mu,\sigma))$ | Log-likelihood function $$\sum_{i=1}^{n} \ln(f(x_i \mid \mu,\sigma))$$ |
| $CDF$ | Cumulative Distribution Function $F$ |
| $PDF$ | Probability Density Function $f$ |

## KOLMOGOROV AXIOMS

$$\mathbb{P}(\Omega)=1$$

$$\mathbb{P}(\varnothing)=0$$

$$\forall A,B\subset\Omega,\ A\cap B=\varnothing.\ (\mathbb{P}(A\cup B)=\mathbb{P}(A)+\mathbb{P}(B))$$

$$\forall A,B\subset\Omega.\ (\mathbb{P}(A\cup B)=\mathbb{P}(A)+\mathbb{P}(B)-\mathbb{P}(A\cap B))$$

## STATISTICALLY INDEPENDENT

$X:\Omega\to\mathbb{R}$ and $Y:\Omega\to\mathbb{R}$ are ***statistically independent***, if the probability density $f_V$ of the random variable

$$V:\begin{cases}\Omega\to\mathbb{R}^2\\ \omega\mapsto\begin{pmatrix}X(\omega)\\Y(\omega)\end{pmatrix}\end{cases}$$

satisfies

$$f_V(x,y)=f_X(x)\cdot f_Y(y)$$

## DISCRETE PROBABILITY DISTRIBUTION

Let $\Omega=\{e_1,e_2,\dots,e_n\}$ be enumerable

$$\mathbb{P}(\{e_k\})=p_k,\quad k\in\{1,2,\dots,n\},\quad E\subset\Omega$$

$$\mathbb{P}(E)=\sum_{e\in E}p(e)$$

$$F(x)=\sum_{e\in\Omega,e\le x}p(e)$$

### *RULES*

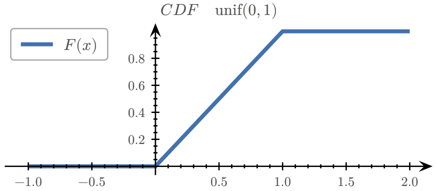
$$\forall e\in\Omega.\ (0\le p(e)\le 1)$$

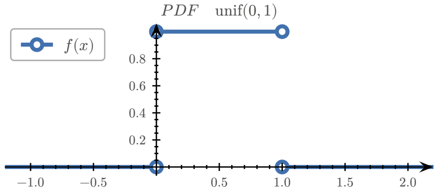
$$\sum_{e\in\Omega}p(e)=1 \qquad \int_{-\infty}^{\infty}f(t)\,\mathrm{d}t=1$$

$$\lim_{x\to-\infty}F(x)=0 \qquad \lim_{x\to\infty}F(x)=1$$

$$0\le F(x)\le 1 \qquad f(t)\ge 0 \text{ for } t\in\mathbb{R}$$

$F(x)$ is monotonically increasing & continuous from the right




## UNIVARIATE UNIFORM DISTRIBUTION

$$X\sim\mathrm{unif}(a,b),\quad\Omega\subset\mathbb{R}$$

$$f_X(x)=\begin{cases}\frac{1}{b-a} & \text{if } x\in(a;b)\\ 0 & \text{else}\end{cases}=\mathbb{1}_{(a;b)}(x)\cdot\frac{1}{b-a}=F_X'$$

$$F_X(x)=\mathbb{P}((-\infty;x]\cap\Omega)=\mathbb{1}_{[a;b]}(x)\cdot\frac{x-a}{b-a}+\mathbb{1}_{(b;\infty)}(x)$$

$$=\mathbb{P}(X\le x)=\int_{-\infty}^{x}f_X(t)\,\mathrm{d}t$$

$$|f_Y(y)\,\mathrm{d}y|=|f_X(x)\,\mathrm{d}x| \quad \text{if monotonically in/decreasing}$$

## UNIVARIATE NORMAL DISTRIBUTION

**Standard normal distribution**:

$$\varphi(x)=\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2}$$

**General normal distribution**:

$$f(x|\mu,\sigma)=\frac{1}{\sigma}\varphi\left(\frac{x-\mu}{\sigma}\right)$$

**Univariate normal distribution**:

$$x\sim\mathcal{N}(\mu,\sigma)$$

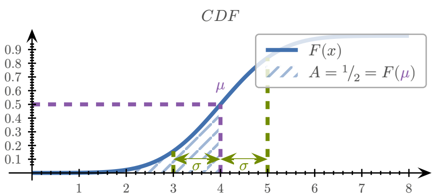
$$f(x|\mu,\sigma)=\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

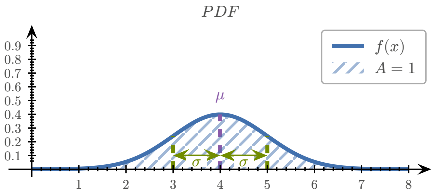
$$\mu\approx\overline{x}=\frac{1}{N}\sum_{i=1}^{N}x_i$$

$$\sigma^2\approx\mathrm{var}=\frac{1}{N-1}\sum_{i=1}^{N}(x_i-\overline{x})^2$$

**Error function**:

$$\mathrm{erf}(x)=\frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\,\mathrm{d}t$$




mean value ($\mu$), standard deviation ($\sigma$)

## MULTIVARIATE NORMAL DISTRIBUTION

$$\Sigma=\text{covariance matrix},\quad V:\Omega\to\mathbb{R}^n$$

$$V\sim\mathcal{N}(\mu,\Sigma)$$

$$f_V(v)=\frac{1}{\sqrt{(2\pi)^n\det\Sigma}}\cdot e^{q_{\Sigma^{-1}}(v-\mu)}$$

$$q_{\Sigma^{-1}}(v-\mu)=-\frac{1}{2}(v-\mu)^\top\Sigma^{-1}(v-\mu)$$

**Sample mean**:

$$\mu\approx\overline{v}=\frac{1}{N}\sum_{i=1}^{N}v_i$$

**Sample covariance matrix**:

$$\Sigma\approx Q=\frac{1}{N-1}\sum_{i=1}^{N}(v_i-\overline{v})(v_i-\overline{v})^\top$$

## EXAMPLES

## CALCULATING LIKELIHOOD FUNCTION

Given the following data and the probability density of the form

| $x$ | $y$ |
| --- | --- |
| 1 | 7 |
| 2 | 16 |
| 3 | 32 |
| 4 | 44 |

$$f(t)=\lambda e^{-\lambda t}\cdot 1_{(0;\infty)}(t)$$

find a formula for the likelihood function:

$$l(\lambda)=f(7)\cdot f(16)\cdot f(32)\cdot f(44)=\lambda^4e^{-99\lambda}$$

Find a formula for the negative log-likelihood:

$$-\ln(l(\lambda))=-\ln\left(\lambda^4e^{-99\lambda}\right)=99\lambda-4\ln(\lambda)$$

## CALCULATING PROBABILITY DENSITY FUNCTION

Given $\sigma,Y$ and $X$, an exponentially distributed random variable with the probability density

$$f_X(x)=X_{[0;\infty)}(x)\cdot e^{-x}$$

$$Y=\sigma(X)$$

$$\sigma(x)=\frac{1}{1+e^{-x}}$$

Calculate the probability distribution function $f_Y(y)$

$$|f_Y(y)\,\mathrm{d}y|=|f_X(x)\,\mathrm{d}x|$$

$$\overset{\frac{\mathrm{d}x}{\mathrm{d}y}>0}{\Leftrightarrow} f_Y(y)=f_X(x)\cdot\frac{\mathrm{d}x}{\mathrm{d}y}$$

$$y=\frac{1}{1+e^{-x}}$$

$$\Leftrightarrow x=-\ln\left(\frac{1}{y}-1\right),\quad 0<y<1$$

$$\Rightarrow\frac{\mathrm{d}x}{\mathrm{d}y}=-\frac{1}{\frac{1}{y}-1}\cdot\frac{-1}{y^2}=\frac{1}{y-y^2}$$

$$\Rightarrow f_Y(y)=X_{[0;\infty)}(x)\cdot e^{-x}\cdot\frac{1}{y-y^2}$$

$$=\frac{1}{y^2}\cdot X_{[0;\infty)}\left(-\ln\left(\frac{1}{y}-1\right)\right)$$

$$-\ln\left(\frac{1}{y}-1\right)\ge 0$$

$$\overset{y>0}{\Leftrightarrow} y\ge\frac{1}{2}$$

$$\Rightarrow X_{[0;\infty)}\left(-\ln\left(\frac{1}{y}-1\right)\right)\overset{y\in(0;1)}{=}X_{[\frac{1}{2};1)}(y)$$

$$\Rightarrow f_Y(y)=\frac{1}{y^2}X_{[\frac{1}{2};1)}(y)$$

***TODO:***
- check if function is increasing ($f'>0$) → Aufgabe 123/124
- distribution to density and vice-versa $X=g(Y)\land g$ increasing $\Rightarrow f_Y(y)=f_X(g(y))\cdot g'(y)$
- FS2024 7.b)
- FS2024 5.b)
- FS2024 6.a)b)

$$\hat{\mu}=\frac{1}{n}\sum_{i=1}^{n}x_i$$

$$\hat{\sigma}=\left(\frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2\right)^{\frac{1}{2}}$$

## WORKING WITH INDICES

$x,y\in\mathbb{R}^n,\ f:\mathbb{R}^n\to\mathbb{R}$

$$f(x)=(x-y)^\top(3x-y)=\sum_{i=1}^{n}(x_i-y_i)(3x_i-y_i)$$

$$\partial x_i f(x)=(3x_i-y_i)+3(x_i-y_i)$$

$$f(x)=\ln(x_1\cdot x_2\cdot\ldots\cdot x_n)-|x|^2=\sum_{i=1}^{n}(\ln(x_i)-x_i^2)$$

$$\partial x_i f(x)=\frac{1}{x_i}-2x_i$$

$$f(x)=\ln(y^\top x)=\ln\left(\sum_{i=1}^{n}y_ix_i\right)\Rightarrow\nabla f=\frac{1}{y^\top x}y^\top$$